GROSS ANATOMY

of the SKULL & BRAIN

A Student-First Practical Identification Guide

Second Semester - Gross Anatomy Practical

What this guide covers

- The scalp, skull terminology, sutures, fontanelles and clinical correlates
- The four norma views + individual cranial and facial bones
- The three cranial fossae and their foramina
- Gross anatomy of the brain - how to identify each part on a specimen
- Real labelled human skull and brain photographs throughout

Table of Contents

This guide reorganises a lecture deck on the gross anatomy of the skull into a logical, exam-ready reference, and adds a complete practical section on the gross anatomy of the brain. Content has been rewritten into full explanations and tables while preserving every anatomical fact, value and clinical correlate from the source material.

How to Use This Guide

This is a **practical-first** study guide. For every structure you are expected to identify on a bone or a brain specimen, the text gives you the **identification cue** - the single most reliable feature to look for - alongside the descriptive anatomy you need for the written paper.

★ How the colour coding works

Crimson bold marks key terms, named structures and values you should be able to recall.

Gold-spined boxes are **study notes** - mnemonics, orientation tips and "how to identify it" cues.

Crimson-spined boxes are **clinical correlates** - the applied anatomy examiners love to ask about.

Navy bars open each numbered section; shaded tables collect comparisons, bones and foramina.

★ Exam orientation tip

Before you name anything, orient the specimen. For a skull, find the **orbits and teeth** (front) and the **foramen magnum** (base). For a brain, find the **frontal pole** (rounded, anterior), the **cerebellum and brainstem** (postero-inferior), and the **longitudinal fissure** (midline). Correct orientation prevents most identification errors.

PART I

Gross Anatomy of the Skull

1 The Scalp

The scalp covers the calvaria (the dome of the skull) and is made of five layers. The first letters spell the word **SCALP**, which is also the standard mnemonic for them. The first three layers are bound tightly together and move as a single unit.

★ Mnemonic - SCALP

S - Skin **C** - Connective tissue (dense) **A** - Aponeurosis **L** - Loose areolar tissue **P** - Pericranium. The outer three (S-C-A) are firmly united and form the mobile scalp proper.

Layer	Description & clinical relevance
S - Skin	Contains sweat glands, sebaceous glands and hair follicles.
C - Connective tissue	Dense and richly vascularised. The fibrous septa hold cut vessels open, so scalp wounds bleed profusely.
A - Aponeurosis	The epicranial aponeurosis (galea aponeurotica). Covers the calvaria and gives attachment to the bellies of occipitofrontalis.
L - Loose areolar tissue	The “danger area” of the scalp - pus or blood spread easily through it, and infection here can pass into the cranial cavity via emissary veins.
P - Pericranium	Forms the external periosteum of the neurocranium (the bones of the cranial vault).

Real human skull, lateral view - the bony surface the scalp covers. | Source: Wikimedia Commons, CC BY-SA 3.0.

⚠ Clinical - the “danger area” of the scalp

Because the loose areolar layer lets blood and pus spread freely, and because **emissary veins** pierce it to connect the scalp veins with the intracranial dural venous sinuses, infection of this layer can be carried into the cranial cavity. This is why the loose areolar layer is called the dangerous area of the scalp.

2 Regional Terminology of the Skull

The skull is described in named regions. Knowing them lets you locate any structure quickly during a practical.

Region	Location / contents
Frontal	Lies in front of the skull (the forehead).
Parietal	Lies on top of the skull; seen from above.
Occipital	Forms the back of the skull.
Temporal	The areas above the ears.
Ocular (orbital)	Contains the eyeballs, the muscles that move the eyes, plus nerves and vessels.
Auricular	Contains the external ears with the external auditory (acoustic) meatus.
Nasal	Contains the external nose and its muscles.
Oral	Contains the upper and lower lips and the tongue. The upper bone is the maxilla, the lower is the mandible.

3 The Skull as a Whole: Cranium & Facial Skeleton

The skull is made of several bones joined together, collectively called the **cranium**. Its single most important task is to protect the brain - the most important organ in the body. The skull is divided into two parts:

Division	Also called	What it is
Neurocranium	The braincase / calvaria ("brain box")	The upper part that encloses and protects the brain.
Viscerocranium	The facial skeleton	The rest of the skull, including the mandible - it forms the face.

★ Key numbers to memorise

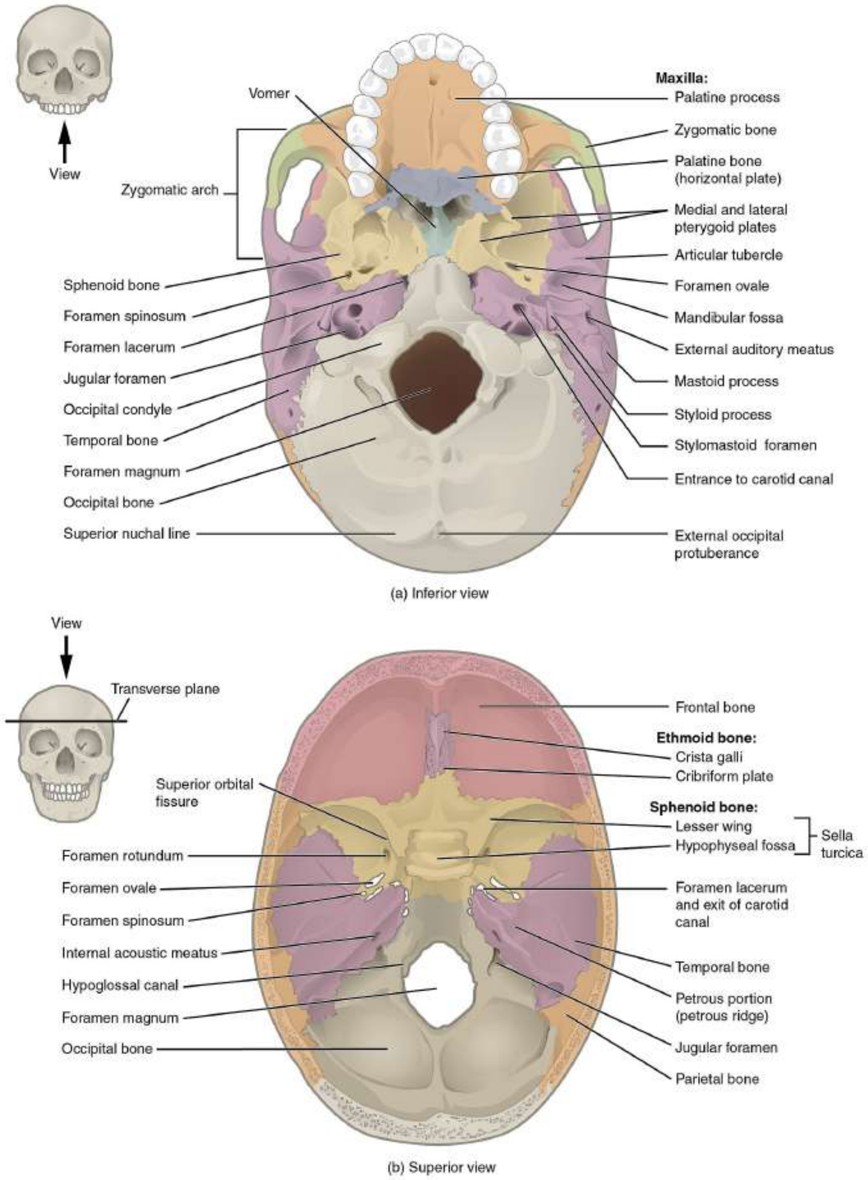
The skull is formed by **28 bones** in total: **14 bones in the calvaria (neurocranium)** and **14 bones in the facial skeleton (viscerocranium)**.

The upper, dome-like part of the skull is the **calvaria** (also spelt calvarium). The facial skeleton constitutes the remainder and includes the freely-moving mandible.

4 Sutures of the Skull

Sutures are the fibrous immovable joints between the flat bones of the skull. They are best appreciated from above (norma verticalis). The main sutures of the adult skull are:

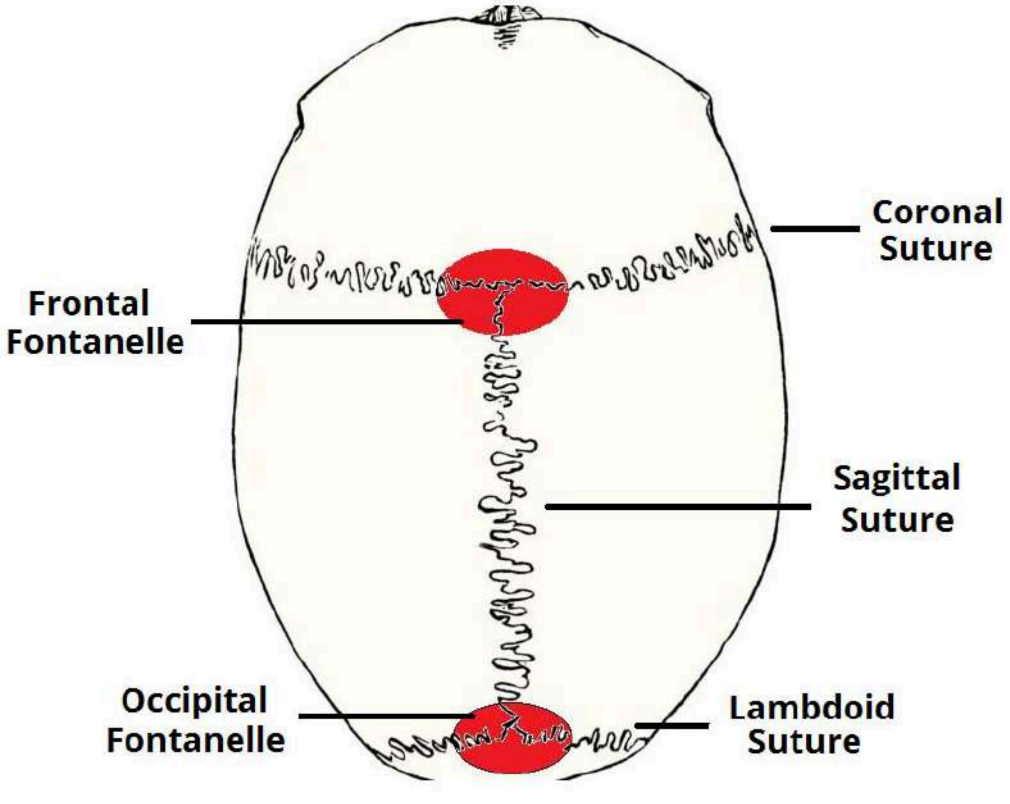
Suture	What it joins
Coronal suture	Fuses the frontal bone with the two parietal bones.
Sagittal suture	Fuses the two parietal bones to each other in the midline.
Lambdoid suture	Fuses the occipital bone to the two parietal bones.
Metopic (frontal) suture	Present in about 3-8% of individuals. It lies in the median plane and separates the two halves of the frontal bone. Normally it fuses by about 6 years of age.



Labelled human skull base (also shows the bones that meet at the sutures). | Source: Wikimedia Commons, CC BY 3.0.

★ How to identify the sutures on a specimen

Hold the skull and look **straight down** on it. The suture running **side-to-side near the front** is the **coronal**; the one running **front-to-back down the midline** is the **sagittal**; the **inverted-V at the back** is the **lambdoid**. Where the sagittal meets the coronal is the **bregma**; where it meets the lambdoid is the **lambda**.



Sutures of the skull seen from above (norma verticalis): coronal, sagittal & lambdoid.

5 Fontanelles of the Neonatal Skull

In neonates the sutures are incompletely fused, leaving membranous gaps between the bones called **fontanelles**. They allow the skull to deform during birth and accommodate rapid brain growth. The two major fontanelles are:

Fontanelle	Fuses at	Adult landmark it becomes
Anterior fontanelle	18 months	BREGMA (junction of sagittal & coronal sutures)
Posterior fontanelle	2-3 months	LAMBDA (junction of sagittal & lambdoid sutures)

Smaller, paired lateral fontanelles also exist: the **sphenoidal (anterolateral)** fontanelles and the **mastoid (posterolateral)** fontanelles, which close in the early months of life.

⚠ Why the anterior fontanelle matters clinically

The anterior fontanelle is the large diamond-shaped soft spot felt on a baby's head. A **bulging, tense** fontanelle suggests raised intracranial pressure; a **sunken** fontanelle suggests dehydration. It is also a site for sampling blood or CSF in infants.

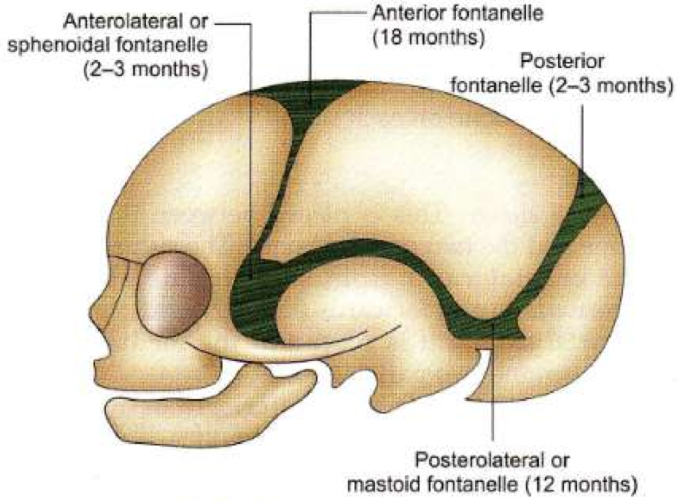


Fig. 1.3: Fontanelles of skull

The neonatal fontanelles: anterior, posterior, sphenoidal and mastoid.

6 Clinical Anatomy: Growth, Moulding & Craniosynosto

Because the sutures and fontanelles are sites of growth, they have several important clinical applications.

Normal growth and birth

The open sutures permit growth of the skull alongside the brain, and their state can help estimate a child's age. During vaginal delivery the calvarial bones can override each other to decrease the size of the head - this is called **moulding** of the calvaria. **Caput succedaneum** is a soft-tissue swelling on the presenting part of the scalp caused by rupture of capillaries during delivery; it resolves within a few days after birth.

If the sutures fuse **too early**, brain growth is restricted: the head growth is stunted and the child may be less intelligent.

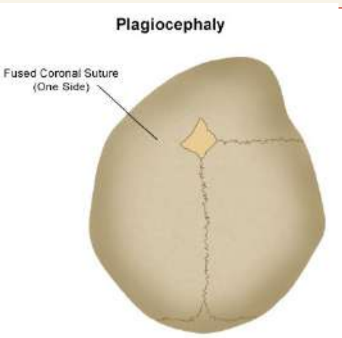
Craniosynostosis - premature closure of sutures

Craniosynostosis is the premature closure of one or more sutures, disturbing normal brain and skull growth. It can raise intracranial pressure and distort the skull and facial bones from their normal symmetrical shape. It features in many genetic syndromes with varied inheritance patterns. The shape of the deformity depends on which suture fuses early:

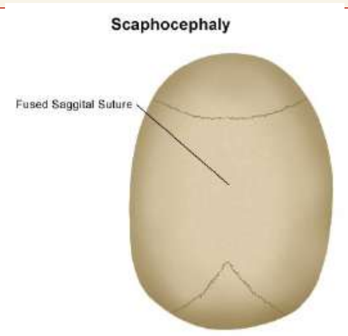
Type	Suture fused early	Resulting head shape
Scaphocephaly	Sagittal suture (front-to-back)	Long, narrow skull - long front-to-back, narrow ear-to-ear.
Anterior plagiocephaly / brachycephaly	One side of the coronal suture (coronal synostosis)	Flattening on the affected side; the forehead and brow on that side stop growing.

⚠ Clinical - extradural (epidural) haemorrhage

A blow to the side of the head at the **pterion** - the thinnest part of the skull - can fracture it and tear the underlying **middle meningeal artery**. Arterial blood then collects between the bone and the dura, producing an **extradural (epidural) haematoma**: a classic lens-shaped (biconvex) collection that does not cross suture lines. Look for a lucid interval followed by deterioration.



Brachycephaly / anterior plagiocephaly (coronal synostosis).

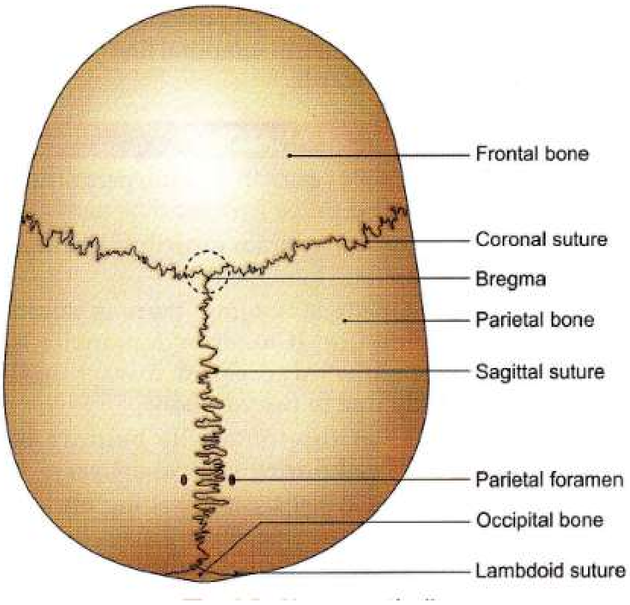


Scaphocephaly (sagittal synostosis).

7 Norma Verticalis (Superior View)

Norma verticalis is the view of the skull **from above**. It is roughly oval, wider behind than in front. It is the best view for studying the sutures and the bones of the vault.

You will see	Identification cue
Frontal bone	The single bone forming the front of the vault.
Two parietal bones	The paired bones forming most of the roof, separated by the sagittal suture.
Occipital bone	Appears at the very back, behind the lambdoid suture.
Bregma	Where the coronal and sagittal sutures meet (former anterior fontanelle).
Lambda	Where the sagittal and lambdoid sutures meet (former posterior fontanelle).
Parietal foramen	A small foramen near the sagittal suture, posteriorly.



Norma verticalis: vault bones, sutures, bregma & lambda.

8 Norma Occipitalis (Posterior View)

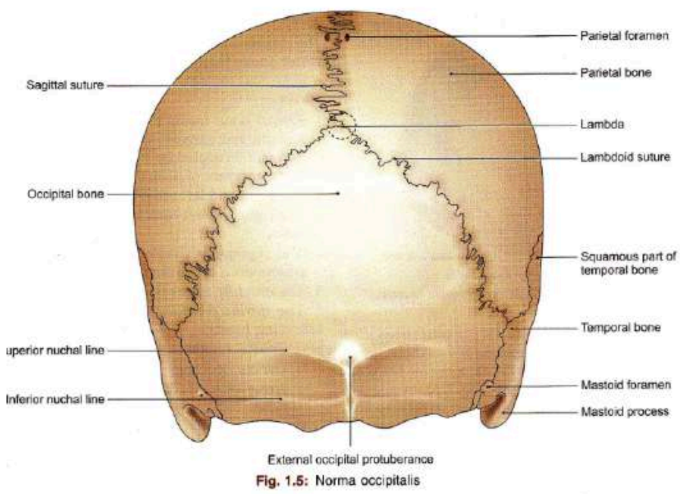
Norma occipitalis is the view of the skull **from behind**. It is convex and shows the occipital bone, the posterior parts of the parietal bones, and the mastoid processes of the temporal bones.

Borders and articulations of the occipital region

It is bordered superiorly and laterally by the **lambdoid suture**, which separates it from the parietal bones. It articulates with the mastoid process of the temporal bone through the **occipitomastoid suture**, while the **petro-occipital suture** joins the petrous part of the temporal bone with the occipital bone. Along these sutures small islands of bone called **wormian (sutural) bones** may be present.

Divisions of the occipital bone

Part	Description
Basilar part	Lies anterior to the foramen magnum and fuses with the sphenoid bone to form the clivus.
Lateral part (x2)	Lies lateral to the foramen magnum; carries the occipital condyle, which articulates with the atlas (C1).
Squamous part	The large flat part behind the foramen magnum. It has an external and an internal surface; the external surface bears the external occipital protuberance (for trapezius) and the superior and inferior nuchal lines.



Norma occipitalis: occipital bone, lambdoid suture & mastoid processes.

9 Norma Lateralis (Lateral View)

Norma lateralis is the **side view** of the skull. It displays both the neurocranium and the facial skeleton and is the key view for the pterion.

Real human skull, lateral view (norma lateralis). | Source: Wikimedia Commons, CC BY-SA 3.0.

★ The bones seen in norma lateralis (mnemonic)

Working roughly front-to-back you can see: **Nasal, Frontal, Lacrimal, Maxilla, Mandible, Zygomatic, Greater wing of Sphenoid, Parietal, Temporal** and **Occipital**. These ten bones are commonly asked to be named in the practical.

⚠ The pterion

The **pterion** is the H-shaped meeting point of the frontal, parietal, sphenoid (greater wing) and temporal bones on the side of the skull. It is the **thinnest part of the skull** and overlies the anterior branch of the **middle meningeal artery** - hence its importance in extradural haemorrhage (Section 6).

10 The Zygomatic Arch & Temporal Fossa

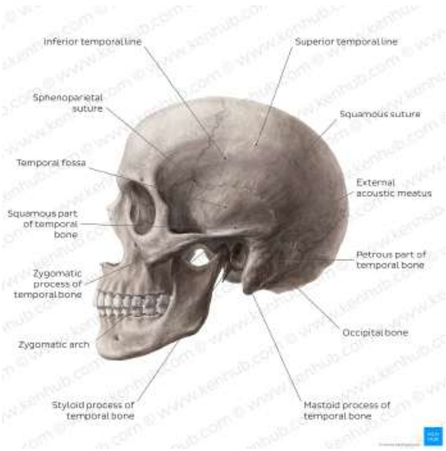
Zygomatic arch

The zygomatic arch is a horizontal bar on the side of the head, in front of the ear. It is formed by the **temporal process of the zygomatic bone** meeting the **zygomatic process of the temporal bone**; the **zygomaticotemporal suture** crosses it obliquely. Above the arch lies the temporal fossa, filled by the temporalis muscle; the **masseter** attaches to its lower margin. The anterior end of its upper border is the **jugal point**, and its posterior end attaches to the squamous temporal bone by anterior and posterior roots.

Temporal fossa

Boundary	Formed by
Above	Superior temporal line.
Below	Upper border of the zygomatic arch.
Laterally	Infratemporal crest of the greater wing of the sphenoid.
Floor	Frontal, parietal, sphenoid (greater wing) and temporal bones - which meet at the pterion.
Contents	The temporalis muscle (which forms the floor) and the pterion.

Bones forming the temporal fossa, summarised: **zygomatic, parietal, frontal, sphenoid and temporal**.



Temporal fossa and its boundaries (norma lateralis).

11 Norma Frontalis (Anterior View) & Facial Bones

Norma frontalis is the **front view** of the skull. It is dominated by the orbits, the nasal aperture and the jaws.

Real human skull, anterior view (norma frontalis). | Source: Wikimedia Commons, CC BY 2.5.

Bone	Contribution to the face
Frontal	Forms the forehead and the superior rim of each orbit.
Nasal (x2)	Form the bridge of the nose.
Maxillae (x2)	Form the upper jaw.
Mandible	Forms the lower jaw.
Zygomatic	Forms the bony prominence of the superolateral cheek; has temporal, frontal and maxillary processes.

Named points of the frontal region

Landmark	How to find it
Superciliary arch	Rounded, curved elevation just above the medial part of the orbit.
Glabella	The smooth median elevation connecting the two superciliary arches.
Frontonasal suture	Below the glabella, where the skull recedes at the root of the nose.
Nasion	The median point where the internasal suture meets the frontonasal suture (root of the nose).

The orbital margins

Margin	Formed by
Supraorbital margin	Frontal bone. The supraorbital notch/foramen sits at the junction of its lateral two-thirds and medial one-third.
Infraorbital margin	Zygomatic bone laterally and maxilla medially.
Lateral orbital margin	Frontal process of the zygomatic bone, completed above by the zygomatic process of the frontal bone.

12 The Maxilla & Mandible

The maxilla

The anterior surface of the maxilla presents the **nasal notch** medially, the **anterior nasal spine**, the **infraorbital foramen** (about 1 cm below the infraorbital margin), the **incisive fossa** above the incisor teeth, and the **canine fossa** lateral to the canine eminence.

The mandible

The mandible forms the lower jaw. Its **upper border (alveolar arch)** lodges the lower teeth; its lower border is rounded. The anterior surface of the body presents the **symphysis menti**, the **mental protuberance** and tubercles, and the **mental foramen**, which transmits the mental nerve and vessels.

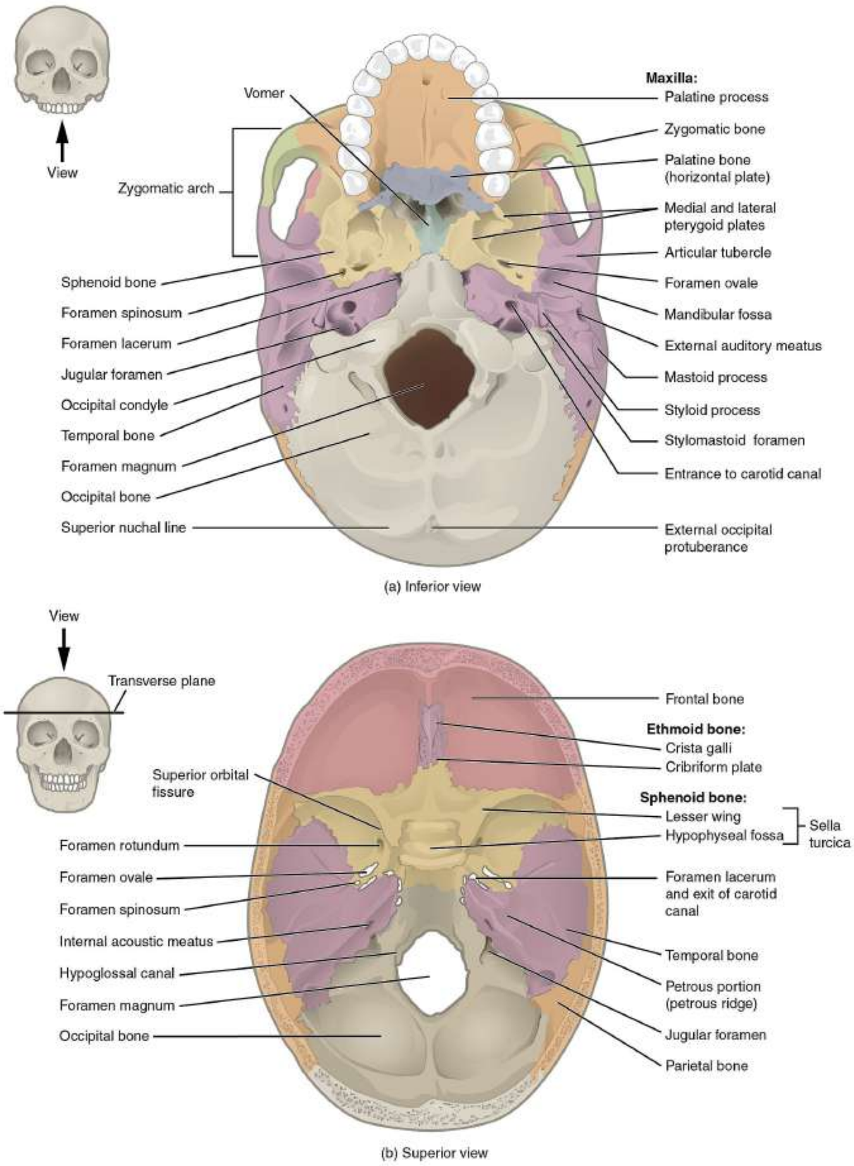
Rami of the mandible

Two **mandibular rami** project perpendicularly upward from the angle of the mandible. Each ramus bears:

Feature	Description / attachment
Head	Situated posteriorly; articulates with the temporal bone to form the temporomandibular joint (TMJ).
Neck	Supports the head; site of attachment of the lateral pterygoid muscle.
Coronoid process	Site of attachment of the temporalis muscle.
Ramus (muscles)	Gives attachment to masseter, temporalis, and medial & lateral pterygoid muscles.

13 Norma Basalis (Inferior View)

Norma basalis is the view of the skull **from below**, with the mandible removed. It is divided into anterior, middle and posterior parts.



Labelled human skull base (norma basalis) and the interior base above. | Source: Wikimedia Commons, CC BY 3.0.

Anterior part - the hard palate

The anterior part forms the **hard palate** and the alveolar arches, which bear sockets for the roots of the upper teeth. The hard palate is formed by the **palatine processes of the maxillae** (anterior two-thirds) and the **horizontal plates of the palatine bones** (posterior one-third). Its sutures are the **intermaxillary, interpalatine** and **palatomaxillary** sutures. The **incisive fossa** lies anteriorly in the midline (pierced by the two incisive canals); the **greater palatine foramen** lies behind the lateral end of the palatomaxillary suture, with the **lesser palatine foramina** behind it.

Middle & posterior parts

The middle part extends from the posterior border of the hard palate to a transverse line through the anterior margin of the foramen magnum. In the midline the **vomer** separates the two posterior nasal apertures (choanae). Laterally lie the **pterygoid processes** and greater wing of the sphenoid, and the three parts of the temporal bone (petrous, tympanic plate and squamous). The posterior part shows the **foramen magnum**, the external occipital crest and protuberance, and the nuchal lines in the midline; laterally lie the condylar and squamous parts of the occipital bone, the **jugular foramen**, the **styloid**

process and the mastoid part of the temporal bone.

14 Interior of the Cranial Base: The Three Fossae

When the skull cap is removed, the floor of the cranial cavity is seen to descend in three steps - the **anterior, middle** and **posterior cranial fossae**. Each lodges a specific part of the brain.

Fossa	Lodges	Key floor bones
Anterior	The antero-inferior parts of the frontal lobes	Frontal bone, cribriform plate of ethmoid, lesser wing of sphenoid
Middle	The temporal lobes (laterally) & pituitary gland (centrally)	Body & greater wing of sphenoid, squamous & petrous temporal
Posterior	The brainstem and cerebellum	Occipital bone, petrous & mastoid temporal, dorsum sellae of sphenoid

Anterior cranial fossa

It lies superior to the nasal and orbital cavities. It is bounded anteriorly and at the sides by the frontal bone, and posteriorly is separated from the middle fossa by the posterior border of the lesser wing of the sphenoid, the anterior clinoid processes and the anterior margin of the sulcus chiasmaticus. In the midline the floor is the **cribriform plate of the ethmoid**, which separates the fossa from the nasal cavity and carries a midline projection, the **crista galli** (which anchors the falx cerebri). On each side of the crista galli the cribriform plate supports the **olfactory bulb** and is pierced by numerous foramina that transmit the olfactory nerve fibres into the nasal cavity, plus the anterior and posterior ethmoidal nerves and vessels.

Middle cranial fossa

Butterfly-shaped, it has a central part formed by the body of the sphenoid that contains the **sella turcica**, which holds the **pituitary gland**; the two lateral parts lodge the temporal lobes. It is bounded anteriorly by the lesser wing of the sphenoid, posteriorly by the superior border of the petrous temporal bone and the dorsum sellae, and laterally by the greater wing of the sphenoid and squamous temporal bone. Its foramina are the **optic canal** and the foramina **rotundum, ovale, spinosum** and **lacerum**.

Posterior cranial fossa

The deepest and largest fossa, it accommodates the **brainstem and cerebellum**. It is bounded anteriorly by the petrous part of the temporal bone and the dorsum of the sphenoid, and posteriorly by the squamous part of the occipital bone; its floor is the basilar and condylar parts of the occipital bone and the mastoid part of the temporal bone. Its great central opening is the **foramen magnum**.

15 Major Foramina of the Skull (Summary)

Foramina are openings that transmit nerves and vessels through the cranial base. This is a high-yield summary table for the practical.

Foramen	Fossa / location	Main structures transmitted
Cribriform plate foramina	Anterior	Olfactory nerve (CN I) fibres
Optic canal	Middle (anterior part)	Optic nerve (CN II), ophthalmic artery
Superior orbital fissure	Middle / orbit	CN III, IV, V1, VI; ophthalmic veins
Foramen rotundum	Middle	Maxillary nerve (CN V2)
Foramen ovale	Middle	Mandibular nerve (CN V3)
Foramen spinosum	Middle	Middle meningeal artery
Foramen lacerum	Middle (base)	Closed by cartilage; internal carotid passes over it
Internal acoustic meatus	Posterior	CN VII (facial) & CN VIII (vestibulocochlear)
Jugular foramen	Posterior	CN IX, X, XI; internal jugular vein
Hypoglossal canal	Posterior	Hypoglossal nerve (CN XII)
Foramen magnum	Posterior	Medulla/spinal cord, vertebral arteries, CN XI (spinal root)

PART II

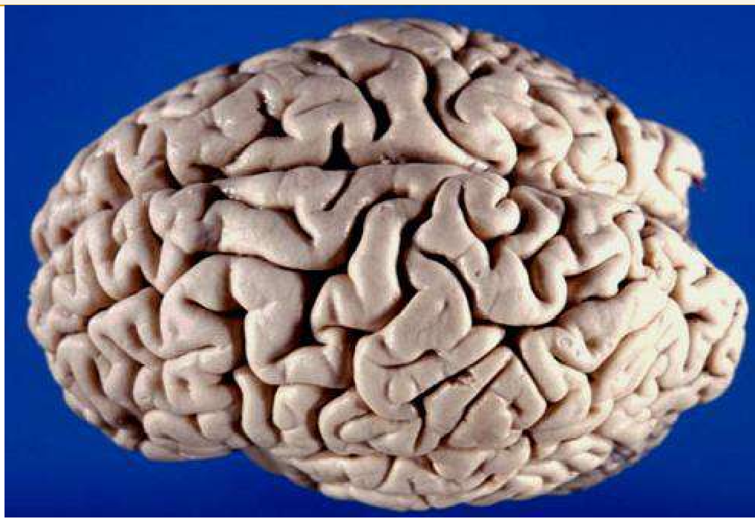
Gross Anatomy of the Brain

16 Orientation & How to Handle a Brain Specimen

The brain has three main parts: the large **cerebrum**, the **cerebellum** behind and below it, and the **brainstem** connecting the brain to the spinal cord. Before naming any structure, orient the specimen correctly.

★ How to orient a whole-brain specimen

1. Find the **two cerebral hemispheres** - the big, wrinkled masses - and the deep midline groove between them, the **longitudinal fissure**. 2. The **frontal poles** (rounded ends) point **anteriorly**; the **occipital poles** point **posteriorly**. 3. Turn it over: the **cerebellum** (tightly folded, “cauliflower” texture) and the stalk-like **brainstem** sit **postero-inferiorly**. 4. The flatter, vessel-covered surface with the cranial nerves is the **inferior (basal) surface**.



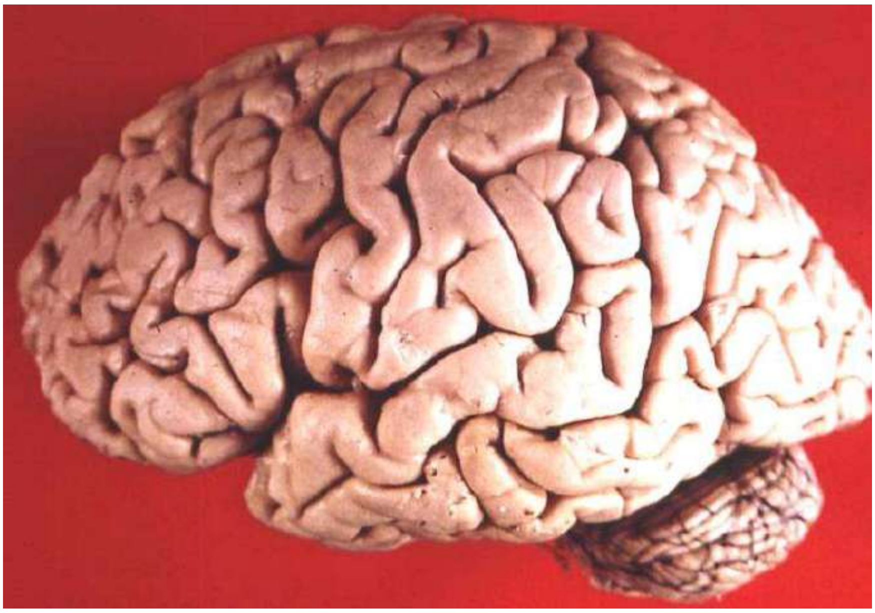
Real human brain, superior view - note the longitudinal fissure dividing the two hemispheres. | Source: Wikimedia Commons, CC BY 2.5.

17 The Cerebrum: Lobes, Sulci & Gyri

The cerebrum is divided into two hemispheres by the longitudinal fissure. Its surface is thrown into folds called **gyri** separated by grooves called **sulci**. Two sulci are the key landmarks for dividing each hemisphere into lobes.

Landmark	How to identify it
Central sulcus	A prominent groove running down-and-forward over the top of the hemisphere; separates the frontal from the parietal lobe. The gyrus in front of it is the motor (precentral) gyrus; behind it the sensory (postcentral) gyrus.
Lateral (Sylvian) fissure	A deep horizontal cleft on the side of the brain; the temporal lobe lies below it.
Parieto-occipital sulcus	Seen mainly on the medial surface; marks the parietal-occipital boundary.

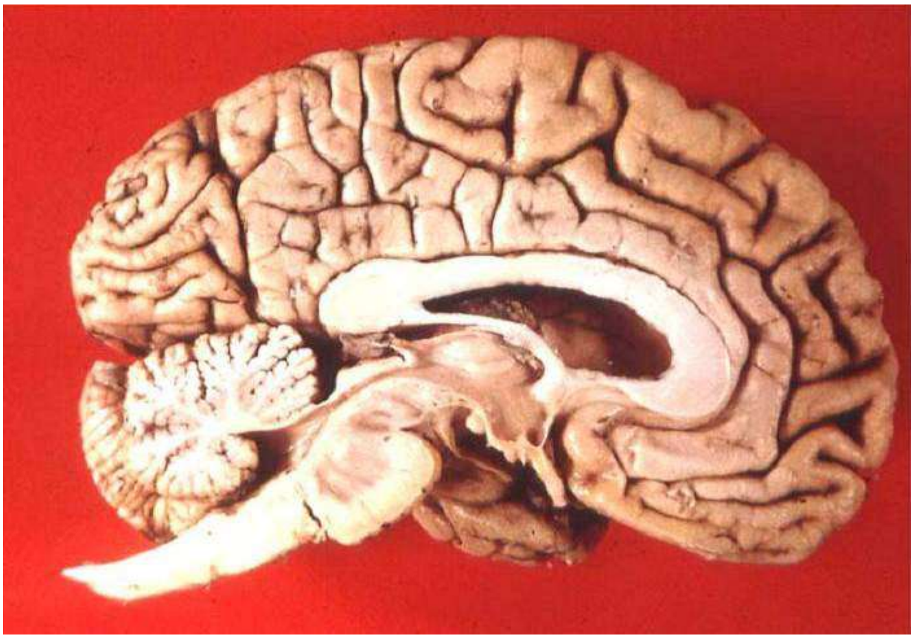
Lobe	Position / identification cue	Chief function
Frontal	In front of the central sulcus, above the lateral fissure	Voluntary movement, personality, speech (Broca)
Parietal	Between central sulcus and parieto-occipital sulcus	General sensation, spatial awareness
Temporal	Below the lateral fissure	Hearing, memory, language (Wernicke)
Occipital	At the very back (occipital pole)	Vision
Insula	Hidden deep within the lateral fissure	Taste, visceral sensation



Real human brain, lateral view - identify the lobes against the central sulcus and lateral fissure. | Source: Wikimedia Commons, CC BY 2.5.

18 The Medial Surface & Midsagittal Landmarks

When the brain is cut in the midline (midsagittal section), the structures connecting and lying between the hemispheres come into view. This is one of the most frequently examined specimens.



Real human brain, midsagittal section - corpus callosum, brainstem and cerebellum are all visible. | Source: Wikimedia Commons, CC BY 2.5.

Structure	How to identify it on the cut surface
Corpus callosum	The large, curved white-matter band arching across the midline - the great commissure connecting the hemispheres. Find it first; everything else is oriented around it.
Septum pellucidum	The thin membrane hanging below the corpus callosum, between the lateral ventricles.
Fornix	A white band curving below the corpus callosum.
Thalamus	The egg-shaped grey mass in the centre, forming the wall of the third ventricle.
Midbrain, pons, medulla	The stepped brainstem descending from the thalamus toward the spinal cord.
Cerebellum	The folded mass behind the brainstem, showing the branching white matter (arbor vitae).

19 The Ventricular System

The ventricles are CSF-filled cavities within the brain. You cannot see them on an intact specimen, but you must know their arrangement and connections.

Ventricle	Location	Connection
Lateral ventricles (x2)	One in each cerebral hemisphere	Drain via the interventricular foramen (of Monro) into the third ventricle
Third ventricle	Midline, between the two thalami	Connects to the fourth via the cerebral aqueduct (of Sylvius)
Fourth ventricle	Between the brainstem (pons/medulla) and cerebellum	Drains into the subarachnoid space via the median & lateral apertures

⚠ Clinical - hydrocephalus

CSF is produced by the choroid plexus, circulates through the ventricles and is absorbed into the venous blood. If flow is obstructed or absorption fails, CSF accumulates and the ventricles enlarge - **hydrocephalus**. In infants (before the sutures fuse) this enlarges the head; in adults it raises intracranial pressure.

20 The Diencephalon

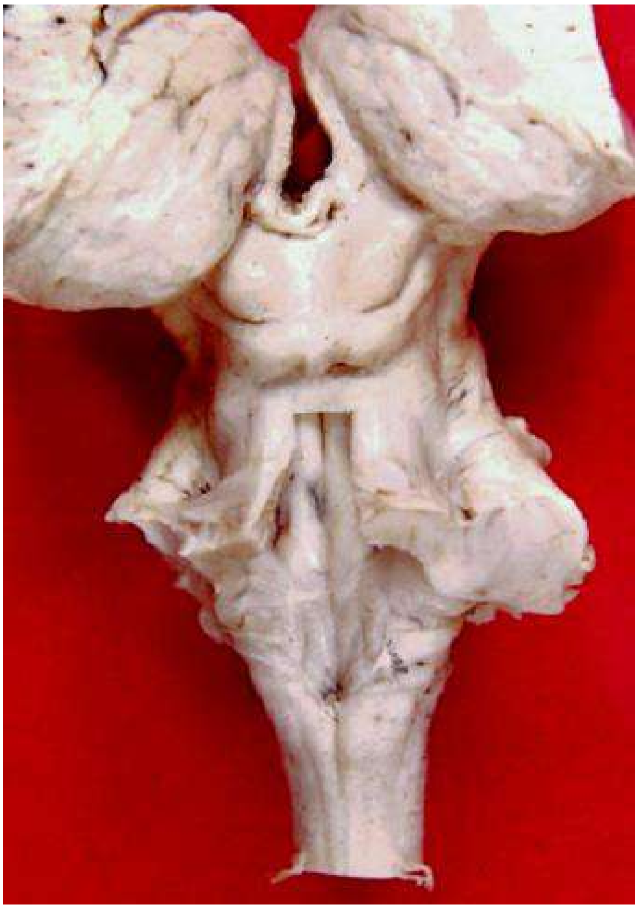
The diencephalon is the central core of the brain, mostly hidden between the hemispheres and best seen on the midsagittal and inferior views. Its main parts:

Part	Identification cue / role
Thalamus	Paired egg-shaped grey masses either side of the third ventricle; the great relay station for sensory information to the cortex.
Hypothalamus	Lies below the thalamus, forming the floor of the third ventricle; controls the autonomic nervous system and pituitary. Identify the mamillary bodies and infundibulum on the inferior surface.
Epithalamus	Posterior; carries the pineal gland.
Pituitary gland	Hangs from the hypothalamus by the infundibulum (stalk); sits in the sella turcica of the sphenoid.

21 The Brainstem: Midbrain, Pons & Medulla

The brainstem connects the cerebrum and cerebellum to the spinal cord and contains the nuclei of cranial nerves III-XII. From above downward it has three parts.

Part	How to identify it	Key surface features
Midbrain	The short uppermost part, just below the thalamus	Two cerebral peduncles in front; four colliculi (corpora quadrigemina) behind
Pons	The bulging middle part	Prominent rounded ventral bulge of transverse fibres; CN V emerges from its side
Medulla oblongata	The lowest part, continuous with the spinal cord	Anterior median fissure with pyramids either side; olives lateral to the pyramids



Real human brainstem with thalamus, posterior view - identify the colliculi and the cerebellar peduncles. | Source: Wikimedia Commons, CC BY 2.5.

★ Quick brainstem identification

On a detached brainstem: the end with the **two thick stalks (cerebral peduncles)** and **four little bumps behind** is the **midbrain (top)**; the **fat rounded belly** is the **pons (middle)**; the tapering end with **two ridges (pyramids)** meeting the cord is the **medulla (bottom)**.

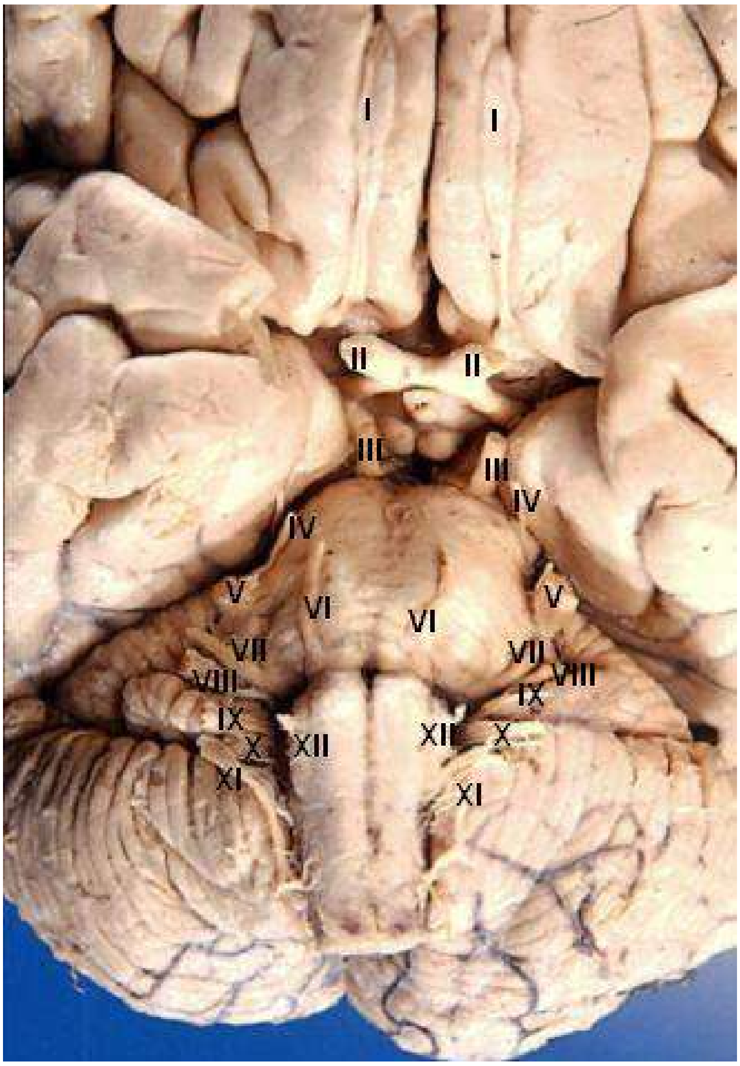
22 The Cerebellum

The cerebellum ("little brain") sits in the posterior cranial fossa, behind the pons and medulla, separated from the cerebrum above by the tentorium cerebelli. It coordinates movement, posture and balance.

Feature	How to identify it
Two cerebellar hemispheres	The paired lateral masses with a tightly folded surface (folia) - finer and more regular than cerebral gyri.
Vermis	The narrow midline ridge connecting the two hemispheres, like a worm.
Folia	The fine parallel surface folds (vs the coarser gyri of the cerebrum) - the best cue that you are holding cerebellum.
Arbor vitae	The branching "tree of life" pattern of white matter seen when the cerebellum is sectioned.
Cerebellar peduncles	Three pairs (superior, middle, inferior) connecting it to the brainstem.

23 The Inferior Surface & The 12 Cranial Nerves

The inferior (basal) surface of the brain is the single most important view for the practical, because all twelve **cranial nerves** are seen emerging here in order. Examiners frequently ask you to identify a numbered nerve on the specimen.



Real human brain, antero-inferior view - the cranial nerves are numbered I-XII as they emerge from the base. | Source: Wikimedia Commons, CC BY 2.5.

★ Mnemonic for the 12 cranial nerves

Names: **O**ld **O**pie **O**ccasionally **T**ries **T**rigonometry **A**nd **F**eels **V**ery **G**loomy, **V**ague **A**nd **H**ypoactive.
Type (motor/sensory/both): **S**ome **S**ay **M**arry **M**oney **B**ut **M**y **B**rother **S**ays **B**ig **B**rains **M**atter **M**ore.

No.	Cranial nerve	Where to find it on the base	Type
I	Olfactory	Olfactory bulbs/tracts on the under-surface of the frontal lobes	Sensory
II	Optic	Form the optic chiasm just in front of the pituitary stalk	Sensory
III	Oculomotor	Emerges from the midbrain, medial to the cerebral peduncle	Motor
IV	Trochlear	The only nerve from the back of the brainstem; thin, curves round the midbrain	Motor
V	Trigeminal	The thick nerve emerging from the side of the pons	Both

No.	Cranial nerve	Where to find it on the base	Type
VI	Abducens	Emerges at the pontomedullary junction, near the midline	Motor
VII	Facial	At the pontomedullary junction, lateral to VI	Both
VIII	Vestibulocochlear	Just lateral to VII at the pontomedullary junction	Sensory
IX	Glossopharyngeal	A row of rootlets from the upper medulla (behind the olive)	Both
X	Vagus	Rootlets from the medulla, below IX	Both
XI	Accessory	Lowest of the rootlet row; has a spinal root ascending through foramen magnum	Motor
XII	Hypoglossal	Rootlets from the medulla between the pyramid and the olive	Motor

Practical Exam Quick-Revision Checklist

Tick these off the night before the practical. If you can do each one on a specimen, you are ready.

Skull

- ☐ Name the five layers of the scalp (SCALP) and the “danger area”.
- ☐ Identify the coronal, sagittal and lambdoid sutures, and locate bregma & lambda.
- ☐ Name the anterior & posterior fontanelles and when they close (18 months / 2-3 months).
- ☐ Identify the four norma views and the bones seen in each.
- ☐ Locate the pterion and explain its clinical importance.
- ☐ Point out the three cranial fossae and what each lodges.
- ☐ Trace a named foramen and state what passes through it.

Brain

- ☐ Orient a whole brain (frontal pole, occipital pole, longitudinal fissure).
- ☐ Identify the central sulcus and lateral fissure and name the four lobes.
- ☐ On a midsagittal cut, find the corpus callosum, thalamus and brainstem.
- ☐ Distinguish midbrain, pons and medulla on a detached brainstem.
- ☐ Recognise the cerebellum by its fine folia and the vermis.
- ☐ Identify any numbered cranial nerve (I-XII) on the inferior surface.

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Human skull base (labelled)	CC BY 3.0	commons: File:707_Superior-Inferior_View_of_Skull_Base-01.jpg
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Human brainstem & thalamus, posterior view	CC BY 2.5	commons: File:Human_brainstem-thalamus_posterior_view.JPG
Human brain, antero-inferior view (cranial nerves)	CC BY 2.5	commons: File:Human_brain_anterior-inferior_view_description.JPG

BRAIN PLATES

High-Yield Specimen Photographs for the Practical

These plates add real human brain specimens for the views examiners set most often at spotter / steeplechase stations. For each, orient first, then work through the structures using the identification cues. All images are openly licensed and credited.

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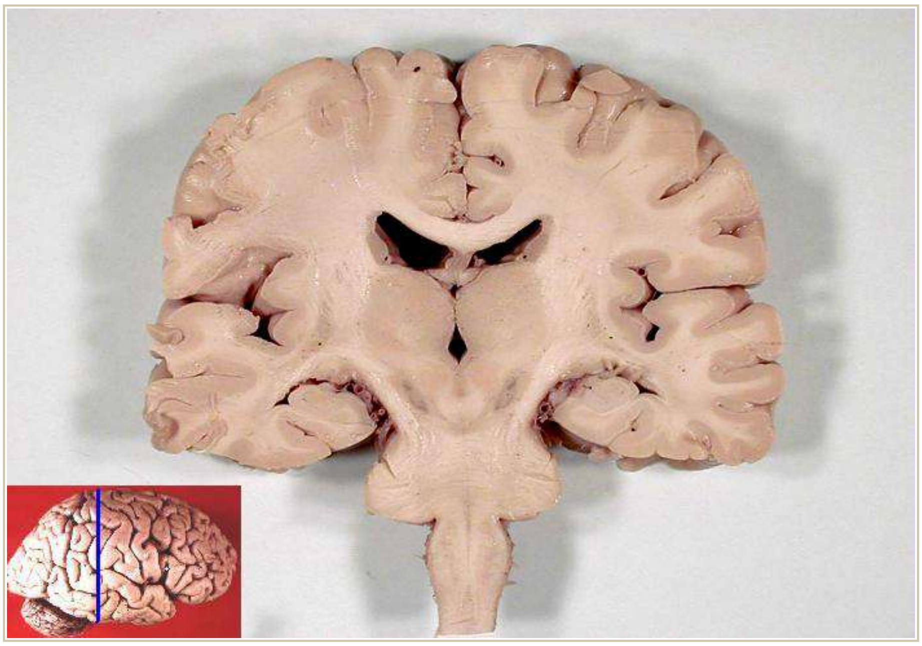
24 The Brain in Coronal Section

A coronal (frontal) cut through the cerebral hemispheres opens up the deep grey and white matter that you cannot see from the surface. This is the standard section for showing the ventricles and the deep grey masses, and is a very common spotter specimen.



Real human brain, coronal section. Identify the cut surface of the cerebral cortex, the central white matter, the lateral ventricles and the deep grey matter. | Source: Wikimedia Commons, John A. Beal PhD, CC BY 2.5.

Structure	How to identify it on the cut surface
Cerebral cortex	The thin grey rim following every gyrus and sulcus around the outside.
White matter (centrum semiovale)	The pale central mass deep to the cortex - myelinated fibres.
Lateral ventricles	The paired slit-like CSF spaces high near the midline.
Corpus callosum	The white band forming the roof of the lateral ventricles, crossing the midline.
Basal ganglia (deep grey)	The grey masses lateral to the ventricles - caudate and lentiform nuclei.
Thalamus	Grey mass forming the wall of the third ventricle (in more posterior cuts).

★ Orientation cue

The longitudinal fissure and the midline run vertically; the two hemispheres are mirror images. Find the ventricles first - everything deep is arranged around them.

25 Arterial Supply: The Circle of Willis

The inferior (basal) surface of the brain carries the arterial circle that supplies it. The Circle of Willis is an anastomotic ring at the base, around the optic chiasm and the pituitary stalk. It is one of the most frequently set arterial spotters.

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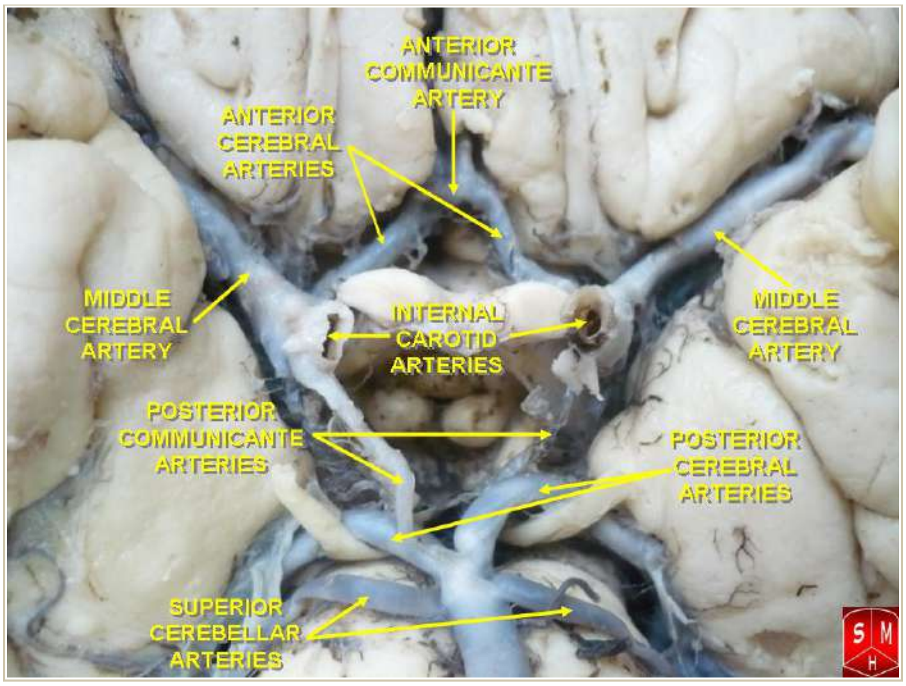

Real human brain, inferior surface - the Circle of Willis with its arteries labelled in situ. | Source: Wikimedia Commons, Anatomist90, CC BY-SA 3.0.

Artery	Where to find it / what it supplies
Internal carotid arteries	Enter at the sides; the main feed to the front of the circle.
Anterior cerebral arteries	Run forward and medially toward the longitudinal fissure.
Anterior communicating artery	The short midline bridge joining the two anterior cerebrals.
Middle cerebral arteries	The large lateral branches passing into the lateral sulcus.
Posterior cerebral arteries	Curve backward; supply the occipital lobe.
Posterior communicating arteries	Join the internal carotid to the posterior cerebral on each side.
Basilar artery	Midline vessel on the pons feeding the back of the circle.

⚠ Clinical - berry aneurysm

The communicating arteries are the classic sites of berry (saccular) aneurysms. Rupture causes a subarachnoid haemorrhage - sudden severe “thunderclap” headache.

26 The Cerebellum (Isolated Specimen)

Removed from the brainstem, the cerebellum shows its surface clearly. Its finely folded surface and central midline ridge make it easy to recognise and a quick, reliable spotter.

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Real human cerebellum, posterior view - note the paired hemispheres, the narrow midline vermis and the fine folia. | Source: Wikimedia Commons, John A. Beal PhD, CC BY 2.5.

Feature	How to identify it
Cerebellar hemispheres	The two large lateral masses with the tightly folded surface.
Vermis	The narrow midline ridge between the hemispheres, like a worm.
Folia	The fine parallel surface folds - finer than the gyri of the cerebrum.
Cerebellar peduncles	Three pairs (cut surface) where it attached to the brainstem.

★ Quick recognition

If the surface folds are very fine and parallel (not the thick wandering gyri of the cerebrum), you are holding the cerebellum.

27 Image Credits - Brain Plates

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Circle of Willis (arterial base)	CC BY-SA 3.0	File:Circle_of_Willis_6.jpg - Anatomist90
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